

Diagnosing Skin Lesions

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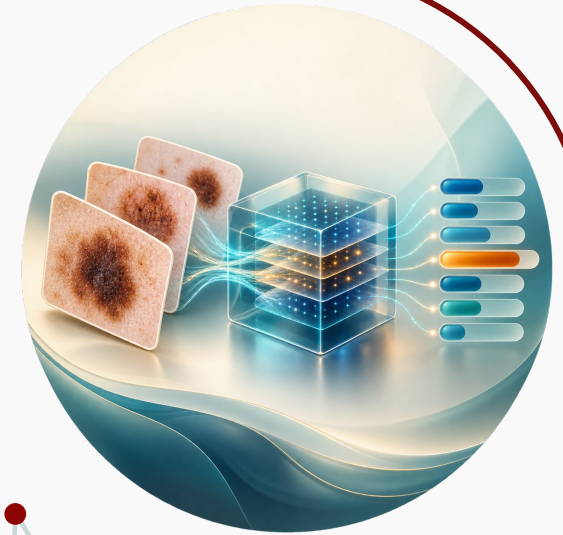
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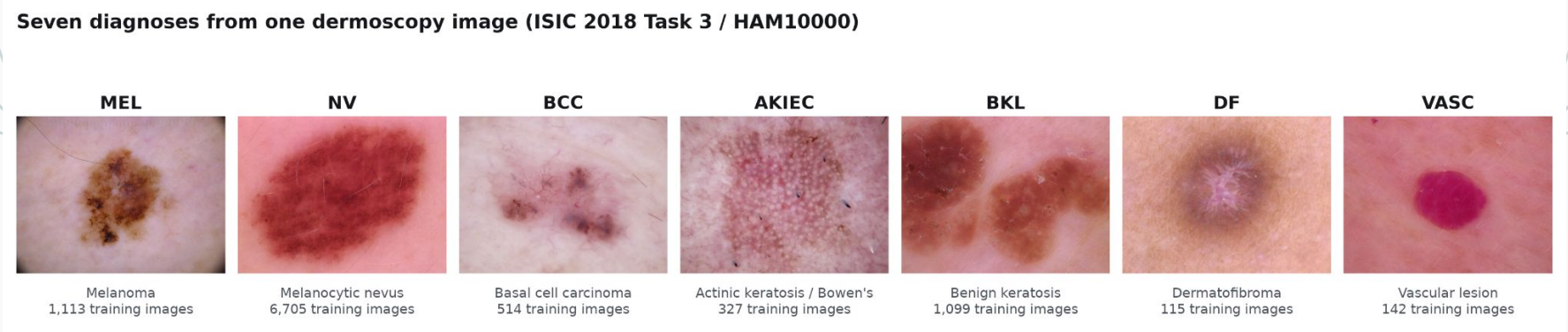
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01. Task



Skin Lesion Diagnosis from One Image

Task: Skin Lesion Diagnosis from One Image



A skin lesion is any spot or growth that differs from the surrounding skin. Dermatologists diagnose it mainly from its appearance under a dermatoscope (shape, border, colours, structures), so a single dermoscopy image is enough for a network to learn the same cues.

Task Ctd.



Importance

Melanoma is curable when caught early. Expert dermoscopy reading is scarce: in the ISIC 2018 reader study, top algorithms scored 25.4 of 30 versus 18.8 for experienced dermatologists.



Applications

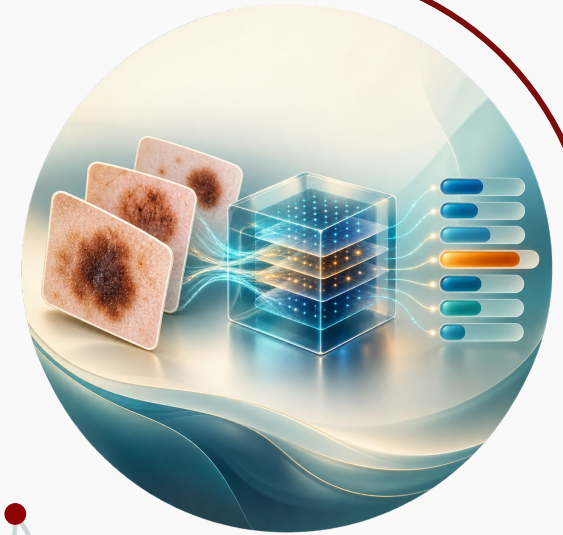
Triage and decision support, teledermatology, primary-care referral decisions and skin-cancer screening programmes.



Goal

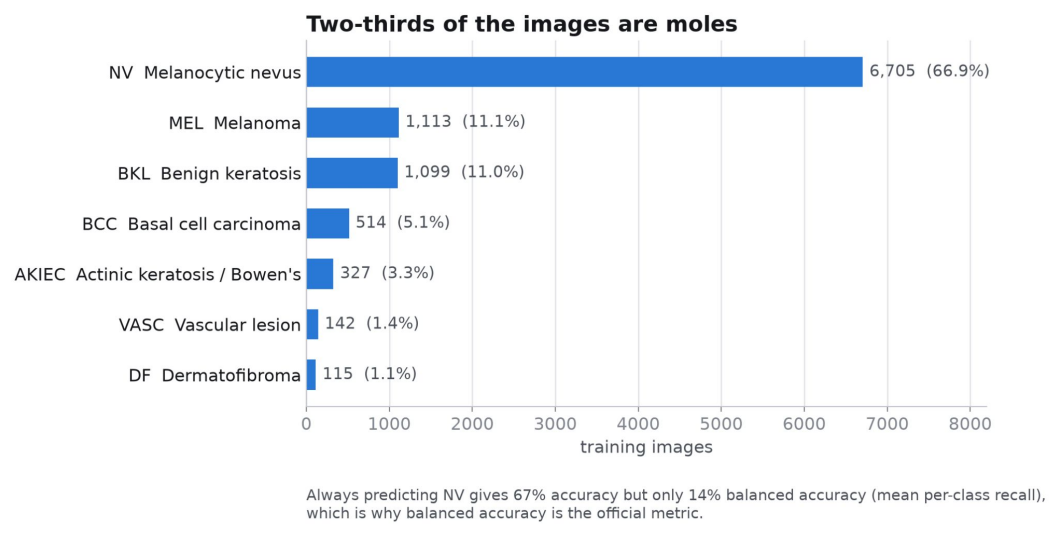
Find which training choices matter, beat a plain baseline and a reproduced published method, and compare with the best published results.

02. Training



Skin Lesion Diagnosis from One Image

Data: HAM10000



10,015

Training images of 7,470 lesions, in 5 folds split by lesion.

1,512

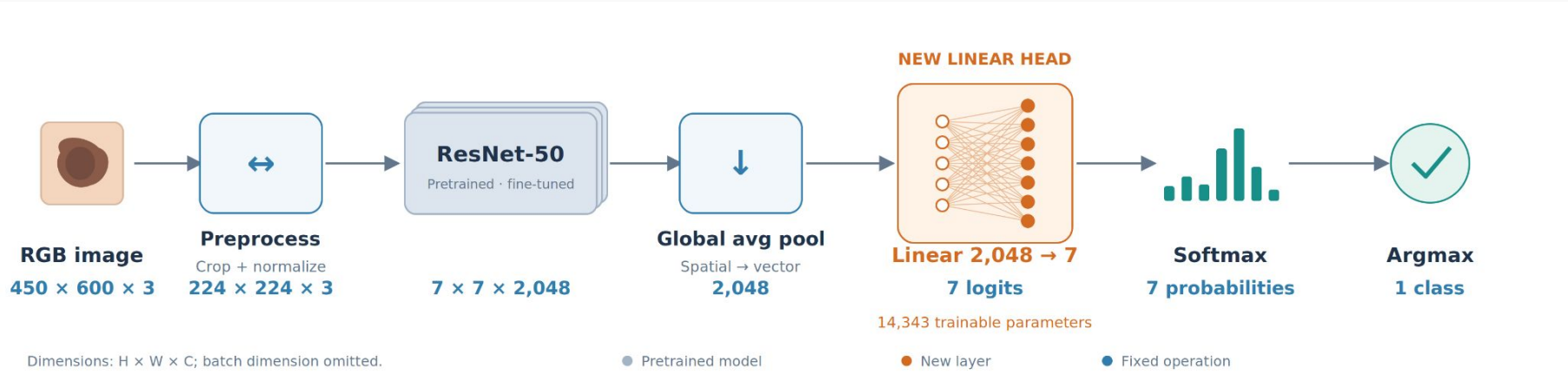
Test images.

Two-thirds of the data are moles, so the data is imbalance. Always guessing “mole” scores 67% accuracy but only 14% balanced accuracy, the official metric (mean of the seven per-class recalls).

Network Summary

	Baseline	Literature Baseline: Gessert et al. 2018	Our Best Model: Distilled ConvNeXt-S
Architecture	ResNet-50, 23.5M params	DenseNet-121, 7.0M params	ConvNeXt-S, 49.5M params
Pretraining	ImageNet-1k	ImageNet-1k	ImageNet-22k
Input	224 px crops	Full-size image, unscaled 224 px crops	448 px crops
Training Targets	True labels, plain loss	True labels, class-weighted loss	Half true labels, half the ensemble's soft labels
Schedule	AdamW, 30 epochs	Adam, 125 epochs	AdamW, 30 epochs
Prediction	One center crop	36 crops averaged	20 views + logit adjustment

Baseline Architecture



The only new layer is the 7-class head (14,343 parameters).
The pretrained ResNet-50 was fine-tuned end to end as well.

Best Model Architecture

Out-of-fold Ensemble Distillation

Step 1 - Teachers

5 Backbones x
5 Folds

ResNet-50, ConvNeXt-S, EfficientNetV2-S, ViT-B/16 and DINOv2, each trained at 448 px on 4 of the 5 folds.

Step 2 - Soft Labels

Out-of-fold Predictions

Every training image gets the calibrated average prediction of the models that never saw it.

Step 3 - Student

One Network

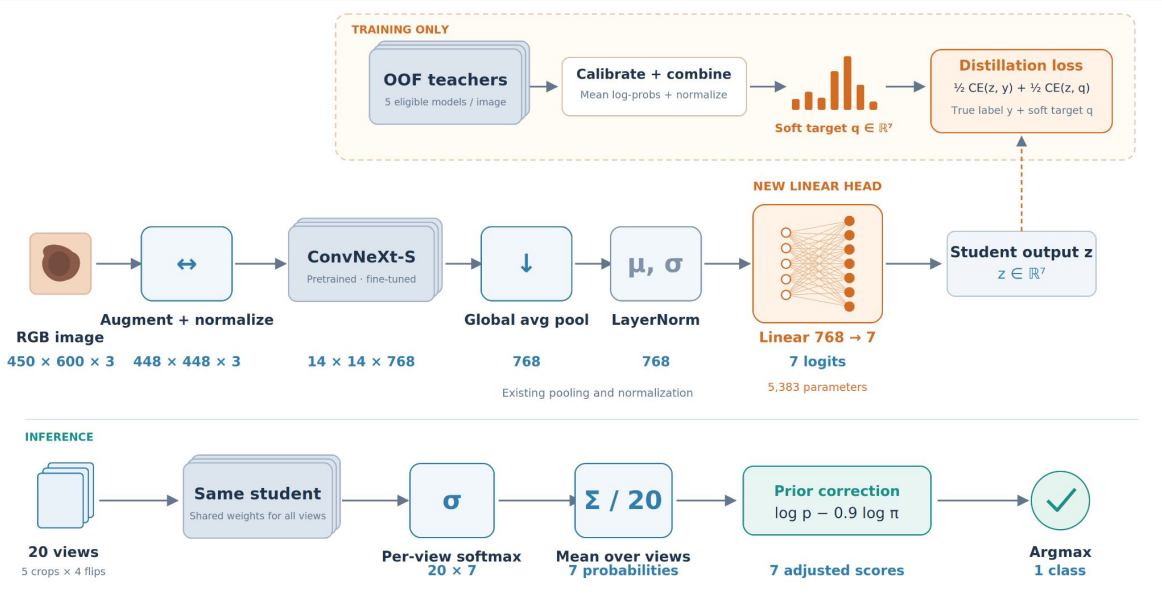
Same method, but the loss is half the true label and half the soft label.

Why it helps: Soft labels say which classes look alike, extra signal where hard labels are thin (only 115 dermatofibromas).

Honest test: The teacher saw each student's validation fold, so students are judged on the test set, settings fixed in advance.

$$\text{Loss} = 0.5 \cdot \text{CE}(\text{true label}) + 0.5 \cdot \text{CE}(\text{soft label})$$

Best Model Architecture Ctd.



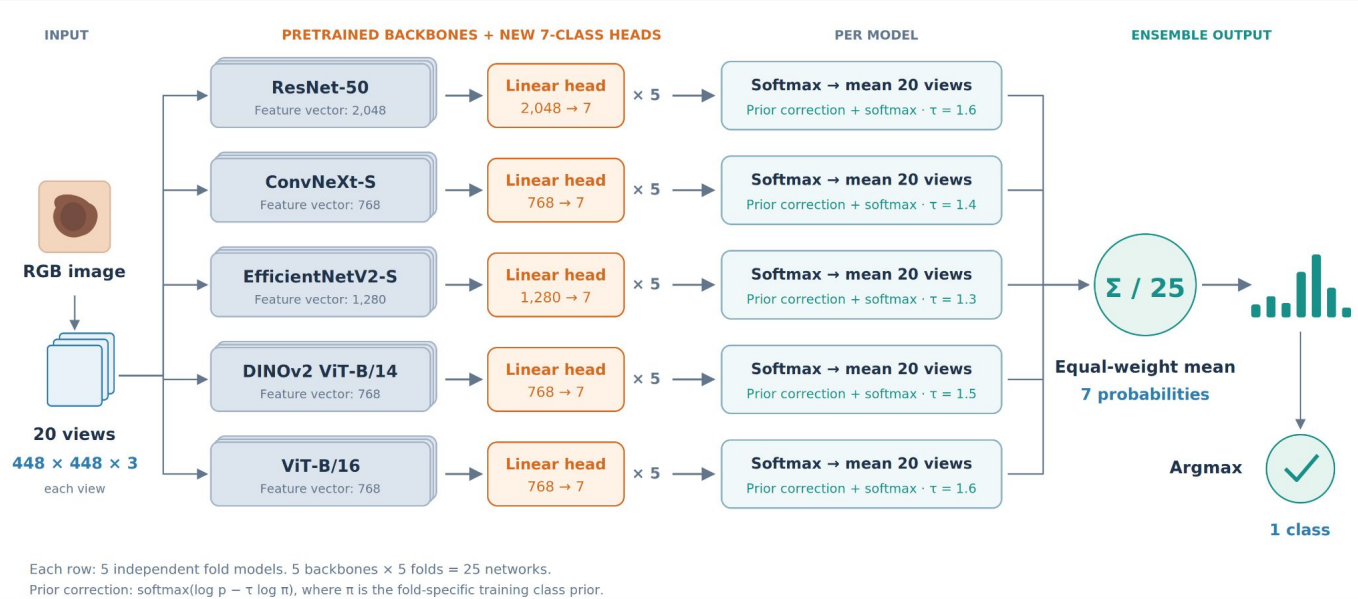
Training

448 px input and a loss that mixes the true label with the out-of-fold ensemble's soft label

Inference

Average 20 views, then subtract τ · log class prior (τ = 0.9, chosen on CV) before the argmax

Our Ensemble Model Architecture



Each backbone's τ was chosen on its own CV predictions. The pre-registered final model uses one calibrated τ over all members instead (0.888).

Training Setup

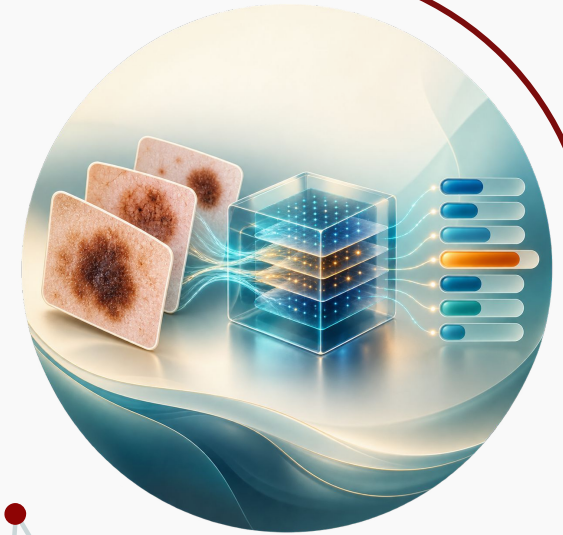
Strategy

- Fine-tune every layer from pretrained weights.
- AdamW, lr 2e-4 for CNNs and 5e-5 for transformers (picked on one fold), batch 64, 30 epochs, warm-up then cosine.
- 448 px random-resized crops, flips, mild colour jitter (bf16).
- Best epoch chosen on the held-out fold.
- Then 20-view test-time averaging and logit adjustment (subtract $\tau \cdot$ log class prior, τ chosen on CV).

Resources

- NERSC Perlmutter, NVIDIA A100 GPUs, one 4-GPU node per round.
- PyTorch and timm pretrained weights used (ISIC 2018 data).
- Gessert et al. 2018 methodology and public code for the literature baseline.

03. Results



Skin Lesion Diagnosis from One Image

Test Results: Balanced Acc.

0.724
Baseline

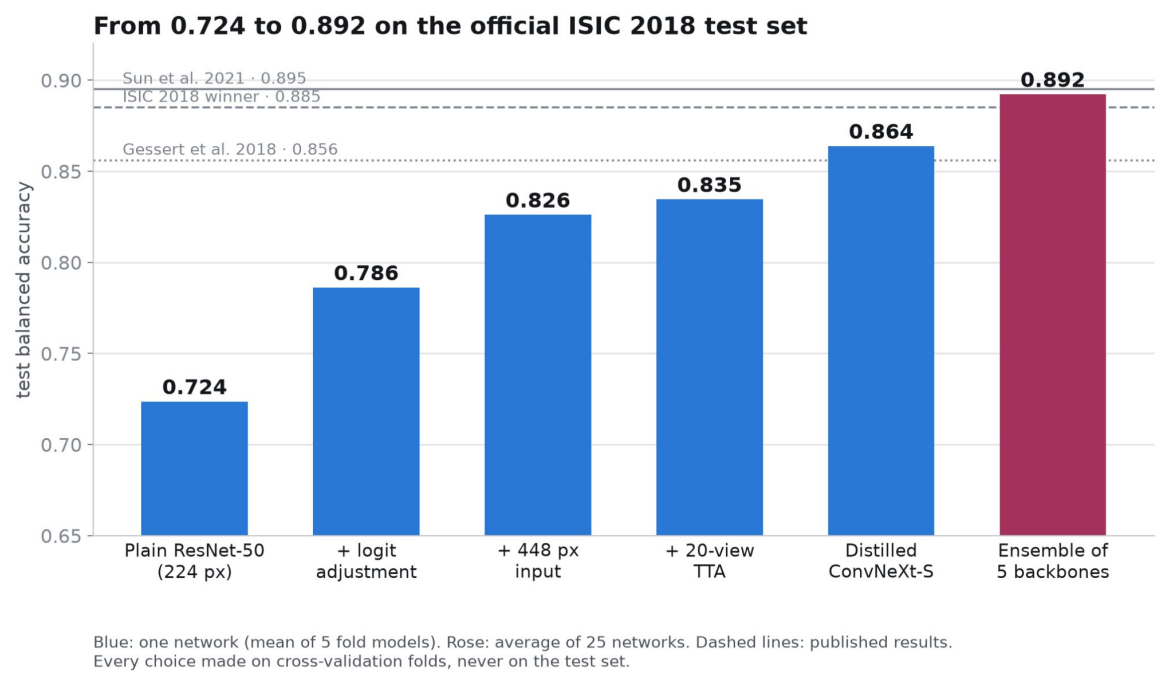
0.770
Literature Baseline

0.864
Our Best Model Distilled ConvNeXt-S

0.892
Our Ensemble Model

Averaging our networks reaches 0.892, a statistical tie with the best published result (0.895), which also used external data and patient metadata.

Path to Best Model



+0.063

Logit adjustment: correct for class imbalance after training, no retraining

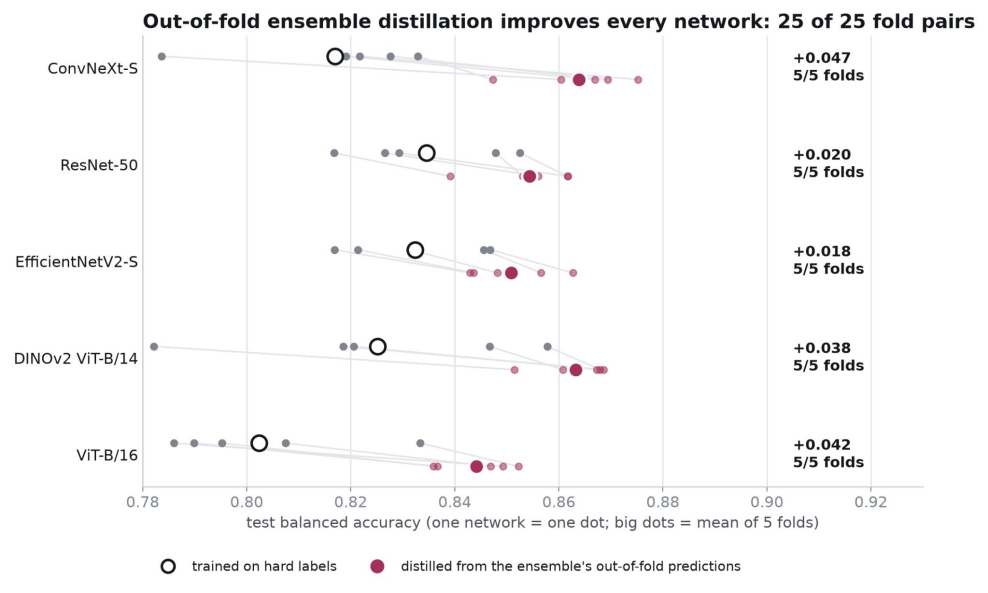
+0.040

448 px input instead of 224 px: finer lesion detail

+0.029

Distillation into ConvNeXt-S, beyond the best ResNet-50 (0.835)

Help of Distillation



25/25

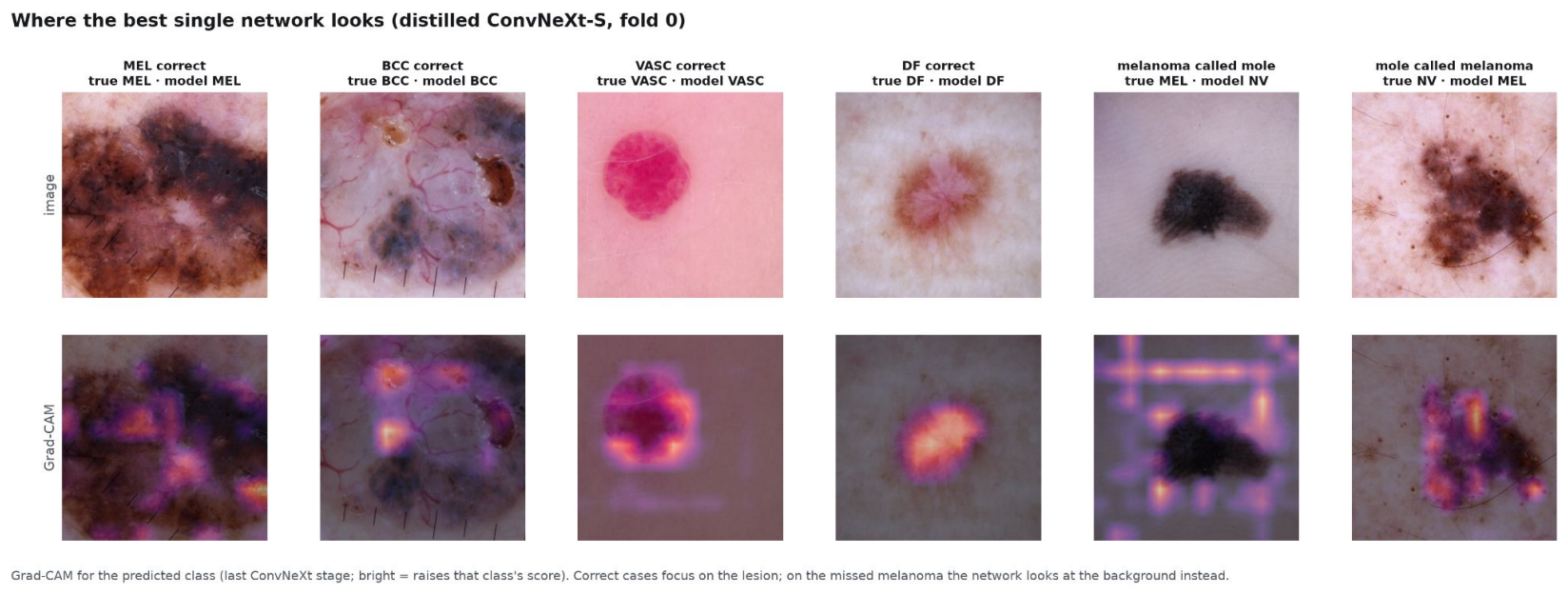
Fold pairs improved, across five architectures.

+0.033

Average gain in test balanced accuracy. AUC also rises on every fold.

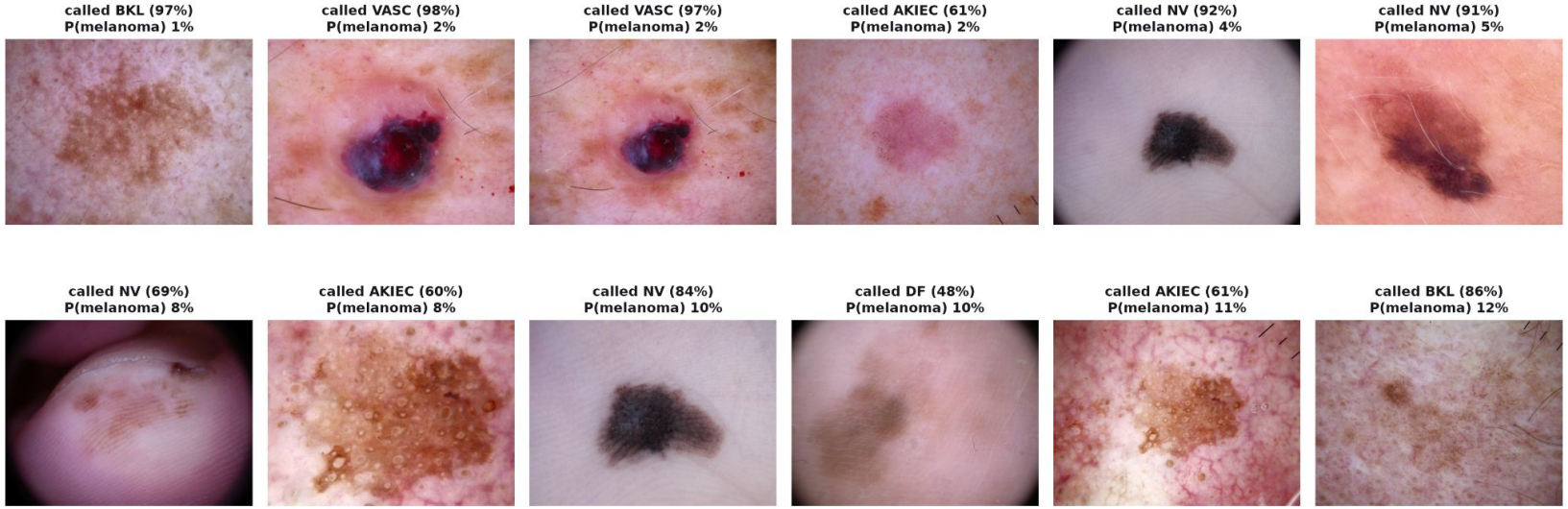
Same recipe, same folds, same seed. Only the training targets change. DINOv2 and ViT-B/16 were a pre-registered replication.

Examples: Model's Behavior



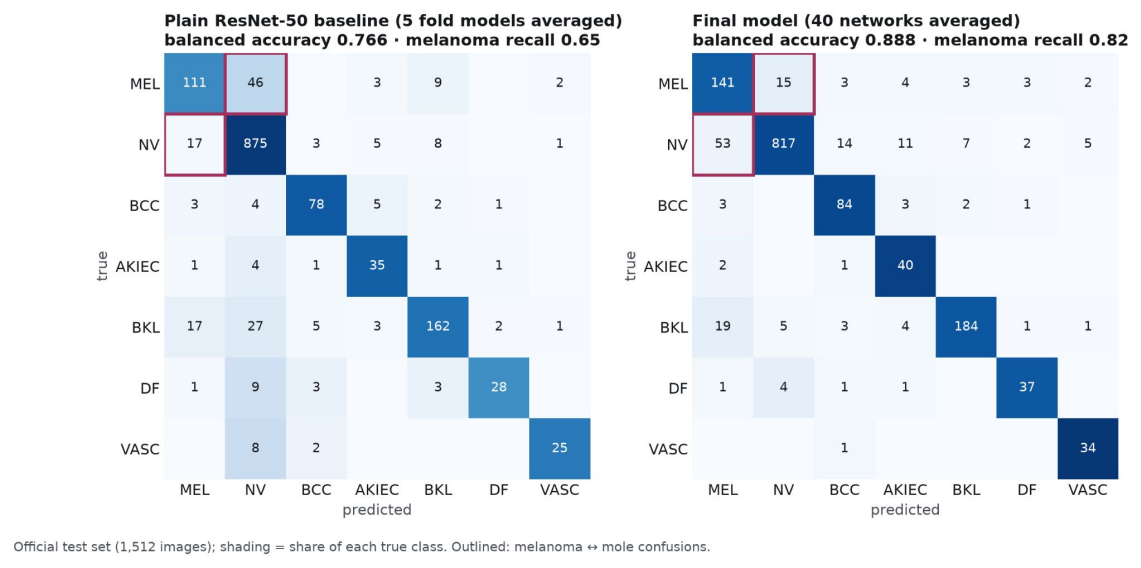
Failure Cases: Melanomas We Missed

Melanomas the final model missed: 30 of 171 (the 12 it was surest about)



Missed melanomas were called: NV 15, AKIEC 4, BKL 3, DF 3, BCC 3, VASC 2. The test set can hold several photos of one lesion (e.g. the two 'VASC' images), so one hard lesion can count twice.

Melanoma-Mole Trade Off



111 → 141

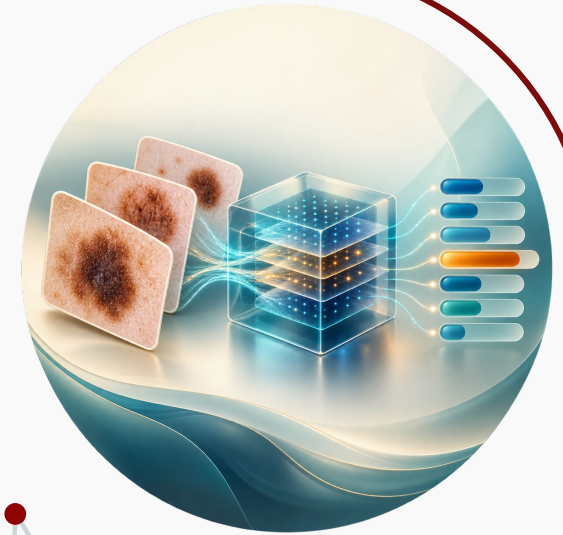
Melanomas caught, of 171

17 → 53

Moles flagged as melanoma, of 909

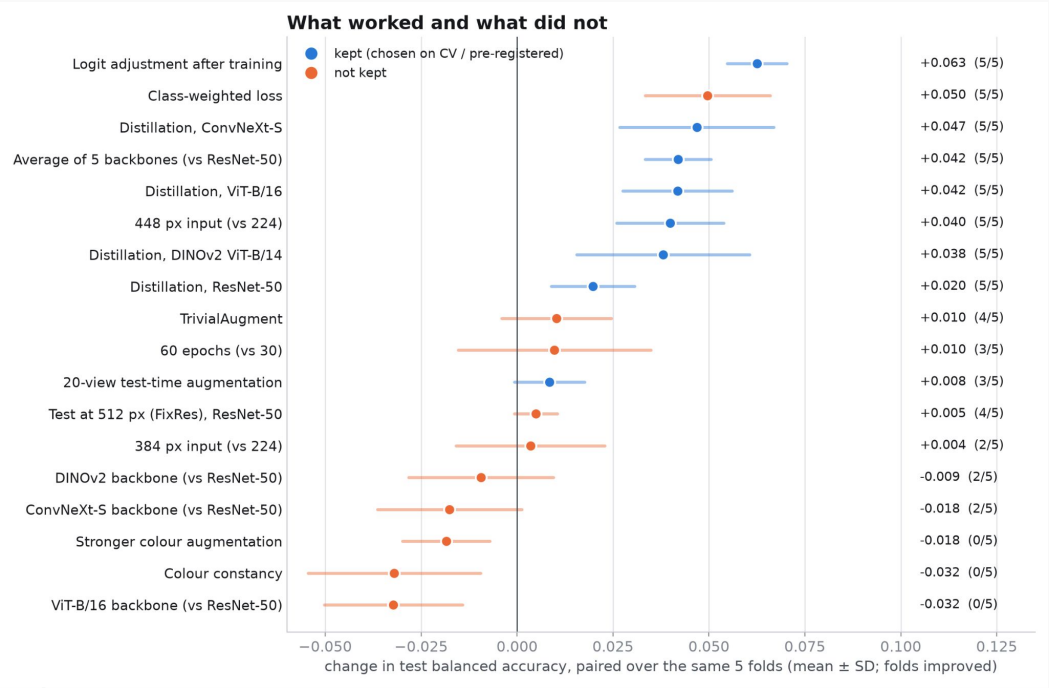
Balanced accuracy rewards this trade.

04. Lessons



Skin Lesion Diagnosis from One Image

Lessons: Methodology Assessment



Worked

Correcting imbalance after training, higher resolution, averaging different backbones, distillation

Did not

A stronger backbone on its own, longer training, extra augmentation, colour normalization, testing at higher resolution

Lessons Ctd.

Learned

- Class imbalance is mostly a decision problem: fixing it after training beat reweighting the loss.
- Resolution and distillation improve the features. AUC rises with them.
- Different architectures make different mistakes, so averaging pays.
- In dermoscopy, color is signal, not noise.

Challenges and Limits

- Test set was too small to separate entries within ±0.025.
- Identical GPU runs differed by 0.025, so we compare 5-fold means, never single runs.
- CV folds come from one source. The test set from several.
- No patient metadata or external data used.

Next

- Fix background attention, for example by cropping to the lesion.
- Medical pretraining (BiomedCLIP).
- More seeds per configuration.
- Calibrated thresholds for a screening setting.

Thank You

Reproducing Gessert et al. 2018

	Paper	Reproduced
5-fold CV balanced accuracy	0.795	0.805
5-fold CV accuracy	0.823	0.847
5-fold CV mean AUC	0.967	0.969
Official test, single model (mean of 5)	-	0.770
Official test, ensemble	0.856 (54 models)	0.787 (5 models)

Same backbone, unscaled crops, class-weighted loss, Adam 5e-4 for 125 epochs, 36-crop evaluation.
Differences: our lesion-grouped folds, ImageNet normalisation, bf16, no 54-model ensemble or SVM stacking.

N. Gessert et al., arXiv:1808.01694 (ISIC 2018 Task 3, 2nd place); code: github.com/ngessert/isic2018.